

SISMO: Split Inhomogeneous Solver for Modelling Oscillations

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ABSTRACT

Context. Oscillation codes that solve the full fourth-order adiabatic problem couple the mechanical variables and the perturbed gravitational potential inside a single operator. A *split* (inhomogeneous) treatment, in which the potential perturbation is computed independently from the current mechanical eigenfunction and injected as a frozen source term, keeps the operator purely mechanical. Such an architecture is attractive as a testbed: additional physics (for instance the Coriolis force) can later be introduced as further source terms without modifying the operator.

Aims. We describe a complete reformulation of the split-Poisson method in which the mechanical problem is reduced to a single scalar second-order equation, modes are located and labelled by a Sturm-type determinant-sign scan with no asymptotic input, and the well-known inaccuracy of split schemes for dipole modes is removed by a parameter-free closure based on Takata's first integral.

Methods. The two mechanical equations of the scaled adiabatic system are reduced exactly to one second-order equation, discretised by a cell-wise exponentially fitted staggered box scheme that yields a symmetrizable scalar tridiagonal matrix. The complete Cowling spectrum is obtained from the parity of a Sturm sequence corrected for the apparent (Lamb-crossing) singularities of the reduction. Each mode is then refined with the frozen-potential source recomputed at every step from an independent Poisson boundary-value problem. For $\ell = 1$ the single free amplitude of that boundary-value problem is fixed by least-squares enforcement of Takata's first integral rather than by the surface matching condition.

Results. On a $1.5 M_{\odot}$ main-sequence benchmark (25 000-point model, $\ell = 1, 2$, $|n| = 1 \dots 100$), the frequencies agree with the fully coupled reference code OSC, run on the identical model, to a median of 2×10^{-5} relative, with a maximum difference of 2.1×10^{-3} over all 198 modes, including the f mode; agreement with GYRE is at the level set by the model re-interpolation ($\approx 2 \times 10^{-3}$). The complete two-degree spectrum is obtained in 6.6 s on 18 threads, about eleven times faster than the coupled reference on the same model.

Conclusions. The gravitational potential can be treated fully inhomogeneously at the accuracy of coupled solvers, provided (i) modes are located on the true resonance structure of the operator rather than on asymptotic estimates, and (ii) the dipole potential is closed by momentum conservation. The architecture is ready for the introduction of rotational source terms.

Key words. asteroseismology – stars: oscillations – methods: numerical

1. Introduction

The computation of linear adiabatic oscillations in a spherically symmetric star is conventionally formulated as a fourth-order eigenvalue problem. The governing system consists of two mechanical equations, the radial motion equation and an equation combining horizontal momentum conservation, mass conservation, and the adiabatic relation, coupled to Poisson's equation for the Eulerian perturbation of the gravitational potential, Φ' (e.g. Unno et al. 1989). Standard stellar oscillation codes, such as OSC (Scuflaire et al. 2008), ADIPLS (Christensen-Dalsgaard 2008), and GYRE (Townsend & Teitler 2013), solve these four equations simultaneously as a single coupled system.

In the context of binary systems and tidally excited oscillations, however, we showed in Fellay & Dupret (2025); Fellay et al. (2026) that retaining Poisson's equation within the coupled system becomes problematic when additional physical effects and mode coupling are included. To overcome this limitation, we developed an iterative approach in which Poisson's equation is solved separately while the mechanical eigenfunctions are held fixed. The resulting gravitational-potential perturbation and its derivative are then introduced as inhomogeneous source terms in the matrix equation governing the mechanical variables

and the mechanical code core is solved again.

Although tidally excited oscillations provide a useful test case for this approach, the same formulation also offers substantial computational advantages for free stellar oscillations. By removing Poisson's equation from the coupled differential system, the discretised mechanical problem can be expressed in terms of a tridiagonal matrix rather than a 4×4 block matrix. This reduction decreases the cost of each matrix factorisation by approximately one order of magnitude. More importantly, the resulting inhomogeneous formulation is readily extensible. Additional physical mechanisms acting on the mechanical variables, most notably the Coriolis acceleration in a rotating star, can be included as source terms evaluated from the current eigenfunction, without requiring the underlying mechanical operator to be reconstructed. The split-Poisson formulation therefore provides a natural framework for such extensions, provided that it can first be demonstrated to reproduce the fully coupled problem with the same level of accuracy.

Achieving this accuracy is not straightforward. Separating Poisson's equation modifies the numerical structure of the standard oscillation problem and requires the formulation of

dedicated discretisation, mode-identification, and iterative-refinement procedures for both the eigenfrequencies and eigenfunctions. In this work, we present a new open-source adiabatic stellar oscillation code, SISMO (Split Inhomogeneous Solver for Modelling Oscillations), based on a split treatment of Poisson's equation, in which the effects not included in the mechanical operator are introduced through inhomogeneous terms.

In Sect. 2, we introduce the principle behind the classical adiabatic oscillation problem, describe our new idea to solve it, and introduce the basic mathematical formalism and variables used in SISMO. Then in Sect. 3 we present the core solver of the code, the discretisation and boundary conditions that are imposed in the code. In Sect. 4, we present the mode finding method and the convergence method used to converge toward the eigenfunctions and eigenvalues. Finally, in Sect. 5, we present benchmark against two classical adiabatic stellar oscillation codes GYRE and CLES, comparing both the frequency and computation times obtained with those codes.

2. Overview of the method

2.1. The problem, and the idea

The linear adiabatic oscillations of a spherical star are described by four coupled first-order equations: two *mechanical* equations, which govern the fluid displacement and the pressure perturbation, and the two components of Poisson's equation, which govern the perturbation of the gravitational potential Φ' and its radial derivative. A coupled code solves all four at once. The idea of the split method is different, we solve only the two mechanical equations, and treat the gravitational potential as a known external forcing, recomputed from the current displacement field between iterations.

This turns the eigenvalue problem into a fixed-point problem. Starting from an approximate eigenfunction, we (i) compute the potential perturbation that this eigenfunction generates, by solving Poisson's equation on its own; (ii) insert that potential into the mechanical equations as a fixed source term; (iii) solve the resulting forced mechanical problem and update the frequency; and (iv) repeat until the eigenfunction regenerates its own potential. At that point the pair (frequency, eigenfunction) solves the original coupled problem exactly: nothing is neglected, the split changes how the solution is found, not what is solved. Its interest is architectural. The operator that must be assembled and factorised at every step contains only the mechanics, so it is small and cheap; and any additional force one wishes to study later (the Coriolis force is our motivation) can be added as one more source term in step (ii), without ever touching the operator. For the case considered here, the split Poisson equation without the Coriolis force, complete calculation proceeds in two stages for each spherical degree ℓ :

1. Locate the modes of the mechanical (Cowling) operator in the requested frequency window. This is done by a counting technique (Sect. 4): a period sweep identifies intervals with an odd change in the corrected count, and bisection locates the enclosed mode. The sampling must be fine enough to resolve adjacent modes.
2. Refine each located mode with the gravitational source switched on (Sect. 4.2), using the fixed-point iteration described above. The frequency moves from its Cowling value to the full value; the shift is below a percent for most modes

and largest for low-order dipole modes, which receive a dedicated treatment (Sect. 4.3).

The remainder of this section defines the variables and the two building blocks (the mechanical pair and the Poisson problem); the numerical core is described in Sect. 3.

2.2. Variables and coefficients

We work with the scaled variables of the OSC adiabatic system 5, which absorb the leading behaviour of the physical quantities at the centre and keep all variables of order unity. With $x = r/R$ the fractional radius and $\sigma^2 = \omega^2$ the squared frequency in units of GM/R^3 , the stellar structure enters through five functions of x :

$$S_2 = \frac{m(x)}{x^3}, \quad S_3 = \frac{P}{\rho}, \quad S_4 = \rho, \quad S_5 = \Gamma_1, \quad S_6 = A_{\text{osc}}, \quad (1)$$

where $m(x)$ is the fractional mass inside x . Two combinations occur repeatedly: the local dimensionless gravity is $g = S_2x$, and the squared Brunt-Väisälä (buoyancy) frequency is $N^2 = S_2S_6x^2$. The four dependent variables are the scaled radial displacement, horizontal displacement, potential perturbation, and potential gradient:

$$\begin{aligned} y_1 &= \frac{\xi_r}{x^{\max(0, \ell-1)}}, & y_2 &= \frac{\sigma^2 \xi_h}{x^{\max(0, \ell-1)} S_2}, \\ y_3 &= \frac{\Phi'}{x^\ell S_2}, & y_4 &= \frac{\partial_x \Phi'}{x^{\max(0, \ell-1)} S_2}. \end{aligned} \quad (2)$$

There is no separate pressure variable: the Eulerian pressure perturbation is carried by the difference $y_2 - y_3$,

$$\frac{P'}{P} = x^\ell \frac{S_2}{S_3} (y_2 - y_3), \quad (3)$$

a relation that follows from the horizontal momentum equation (which is algebraic in the adiabatic case) and that will be needed again for the dipole closure of Sect. 4.3. In these variables the two mechanical equations take the compact form

$$\frac{dy_1}{dx} = a_{11}y_1 + a_{12}y_2 + a_{13}y_3, \quad (4)$$

$$\frac{dy_2}{dx} = a_{21}y_1 + a_{22}y_2 + a_{23}y_3, \quad (5)$$

with the coefficients

$$\begin{aligned} a_{11} &= -\frac{\ell+1}{x} + d_{13}, & a_{12} &= \frac{\ell(\ell+1)S_2}{x\sigma^2} - d_{13}, \\ a_{13} &= d_{13}, & a_{21} &= c_{21} + \frac{\sigma^2}{xS_2}, \\ a_{22} &= c_{33} - c_{21}, & a_{23} &= c_{21}, \end{aligned} \quad (6)$$

where

$$d_{13} = \frac{xS_2}{S_3S_5}, \quad c_{21} = xS_6, \quad c_{33} = \frac{3 - S_4/S_2 - \ell}{x}. \quad (7)$$

Note that the potential enters the mechanics only through y_3 (via a_{13} and a_{23}); the gradient y_4 does not appear. This is why, in the split scheme, only y_3 needs to be supplied as a source. One coefficient deserves a warning, because the point is invisible in the usual block formulation. In the underlying OSC system the

derivative of the first equation carries a weight σ^2 ; once the equation is divided by that weight, the off-diagonal coefficient a_{12} acquires the term $\ell(\ell+1)S_2/(x\sigma^2)$. This $1/\sigma^2$ is not a detail: it is what carries the gravity-mode physics. A quick way to see it is to form the local radial wavenumber of the pair,

$$k_r^2 \simeq -a_{12} a_{21} \simeq \frac{\ell(\ell+1)}{x^2} \left(\frac{N^2}{\sigma^2} - 1 \right), \quad (8)$$

which is the standard asymptotic dispersion relation of gravity modes, with its unbounded accumulation of modes as $\sigma \rightarrow 0$. An implementation that forgets the derivative weight loses the $1/\sigma^2$ and produces a spectrum with only a finite number of gravity modes – a qualitatively wrong result that is nevertheless easy to mistake for a subtle bug. Equation (8) is a one-line sanity check that we recommend to anyone re-implementing the system.

2.3. The independent Poisson problem

The potential pair obeys the two remaining equations of the system,

$$\frac{dy_3}{dx} = c_{33} y_3 + \frac{1}{x} y_4, \quad (9)$$

$$\frac{dy_4}{dx} = -c_{21} \frac{S_4}{S_2} y_1 + c_{42} y_2 + \left[\frac{\ell(\ell+1)}{x} - c_{42} \right] y_3 + \left(c_{33} - \frac{1}{x} \right) y_4, \quad (10)$$

with $c_{42} = xS_4/(S_3S_5)$. These are the scaled form of $\nabla^2\Phi' = 4\pi G\rho'$: the terms in y_1 and y_2 on the right-hand side of Eq. (10) are the density perturbation generated by the displacement field. In the split scheme these equations never enter the eigenvalue operator. They define a small, self-contained boundary-value problem: given the current mechanical iterate (y_1, y_2) , find the potential it generates.

The problem is linear, so its solution is built from two pieces. A particular solution (y_3^p, y_4^p) is integrated outward from the centre (fourth-order Runge–Kutta) with the (y_1, y_2) source, it contains the response to the density perturbation, but does not yet satisfy the outer boundary condition. A homogeneous solution (y_3^h, y_4^h) is integrated starting from the regular central behaviour $y_4 = \ell y_3$. The full solution is

$$\begin{pmatrix} y_3^\Phi \\ y_4^\Phi \end{pmatrix} = \begin{pmatrix} y_3^p \\ y_4^p \end{pmatrix} + A \begin{pmatrix} y_3^h \\ y_4^h \end{pmatrix}, \quad (11)$$

where A is a constant classically set by the vacuum condition at the surface (the interior potential must match a decaying exterior solution),

$$y_4 + (\ell+1)y_3 = -\frac{S_4}{S_2} y_1 \quad (x = x_s). \quad (12)$$

For $\ell \geq 2$ this closure works well. For the dipole it can become problematic. The homogeneous solution grows outward as x^ℓ , through the envelope and the model atmosphere; the amplitude A is fixed by a condition imposed at the very top of that growth; and the resulting potential is then used deep in the interior, where, for $\ell = 1$ specifically, it has a strong effect on the frequency. A small inaccuracy at the surface is thus propagated globally and amplified. Sect. 4.3 replaces this closure for $\ell = 1$ by a condition that involves the whole star instead of one point.

3. The scalar second-order core

3.1. From two equations to one

Because the gravitational-potential perturbation is held fixed during each mechanical solve, Eqs. (4)–(5) form a linear system of two first-order differential equations with a prescribed source term. This system can be reduced exactly to a single second-order equation for either mechanical variable. Focussing on y_1 , which is directly related to the displacement, and eliminate y_2 using Eq. (4) gives:

$$y_2 = \frac{y_1' - a_{11}y_1 - a_{13}y_3}{a_{12}}. \quad (13)$$

Substitution into Eq. (5) yields a second-order equation of Sturm–Liouville type,

$$y_1'' + \alpha(x)y_1' + \beta(x)y_1 = R(x), \quad (14)$$

where the coefficients α and β are combinations of the a_{ij} and their derivatives, while the source term R depends linearly on y_3 and its derivative. This reduction introduces no approximation. Its main advantage is numerical: discretising a scalar second-order equation on N mesh points produces an $N \times N$ tridiagonal system, instead of the 4×4 block-banded system associated with the original formulation. The resulting linear system can therefore be factorised and solved at substantially lower computational cost.

The reduction nevertheless introduces a division by a_{12} in Eq. (13). This coefficient vanishes at the radius where σ^2 equals the local Lamb-frequency term,

$$\sigma^2 = \frac{\ell(\ell+1)S_3S_5}{x^2}. \quad (15)$$

This apparent singularity does not compromise the formulation for two reasons. First, the choice to eliminate y_2 , rather than y_1 , places the corresponding singular point in a comparatively benign region. For gravity modes, the Lamb-frequency crossing generally lies in the outer evanescent region, away from the main propagation cavity. Eliminating y_1 instead would introduce a singularity where $\sigma^2 = N^2$, namely close to the turning points that delimit the propagation cavity of each g mode.

Second, the singularity is not present in the underlying first-order system, which remains regular at the crossing. The two local solutions of Eq. (14) also remain regular. The vanishing of a_{12} therefore represents a local differential constraint rather than a physical singularity. At the crossing radius, Eq. (4) loses its dependence on y_2 and directly relates y_1' to y_1 and to the prescribed source. A suitable discretisation must recover this constraint without requiring a special matching condition. As shown below, this property is built into the adopted numerical scheme.

3.2. Discretisation on a node–face grid

Section 3.1 shows that the two mechanical equations can be reduced to one second-order equation for y_1 . In the code, we construct the discrete scalar equation directly from the original first-order pair. This avoids differentiating the stellar-model coefficients and preserves the regular behaviour of the original equations when a_{12} vanishes.

We use the original stellar grid without remeshing. The displacement variable y_1 is stored at the model points x_i , called

nodes, whereas y_2 is stored halfway between adjacent nodes, at faces. Around node i , the arrangement is

$$\underbrace{y_{1,i-1}}_{\text{node}} \longleftrightarrow \underbrace{y_{2,f^-}}_{\text{face}} \longleftrightarrow \underbrace{y_{1,i}}_{\text{node}} \longleftrightarrow \underbrace{y_{2,f^+}}_{\text{face}} \longleftrightarrow \underbrace{y_{1,i+1}}_{\text{node}}.$$

This placement follows the equations themselves. Equation (4) describes the change of y_1 across one cell, so it connects two nodes through the face value of y_2 . Equation (5) describes the change of y_2 across the control volume around a node, so it connects the two neighbouring faces through the nodal value of y_1 .

Within each cell, the coefficients are treated as constant and the differential equation is integrated analytically. Applying this construction to Eq. (4) over the cell $[x_i, x_{i+1}]$, of width h , gives

$$y_{1,i+1} = e^p y_{1,i} + h \varphi_1(p) (\bar{a}_{12} y_{2,f} + \bar{a}_{13} \bar{y}_3), \quad p = \bar{a}_{11} h, \quad (16)$$

where overbars denote cell averages and

$$\varphi_1(p) = \frac{e^p - 1}{p}, \quad \varphi_1(0) = 1.$$

The factor φ_1 makes the update exact for the piecewise-constant coefficients used in the cell. For small p , it tends to the usual finite-difference update. More importantly, Eq. (16) can be read simply as follows: given y_1 at the two nodes and the source y_3 , it determines the value of y_2 at the face between them.

Applying the same construction to Eq. (5) over the volume between faces f^- and f^+ , of width V_i , gives

$$y_{2,f^+} = e^q y_{2,f^-} + V_i \varphi_1(q) (a_{21,i} y_{1,i} + a_{23,i} y_{3,i}), \quad q = a_{22,i} V_i. \quad (17)$$

We now eliminate the face values. Equation (16) expresses y_{2,f^-} in terms of $y_{1,i-1}$ and $y_{1,i}$, and y_{2,f^+} in terms of $y_{1,i}$ and $y_{1,i+1}$. Substituting these two expressions into Eq. (17) leaves one equation involving only three neighbouring values of y_1 . Repeating this at every node gives

$$\mathbf{T}(\sigma^2) \mathbf{y}_1 = \mathbf{r}(y_3), \quad (18)$$

where $\mathbf{T}$ is tridiagonal and $\mathbf{r}$ contains the prescribed gravitational source. This is the discrete scalar problem introduced in Sect. 3.1; the face variable y_2 is reconstructed only when it is needed.

The Lamb-frequency crossing. Solving Eq. (16) for $y_{2,f}$ introduces a factor $1/\bar{a}_{12}$. The coefficients therefore become large near the radius where $\bar{a}_{12} = 0$, but the physical solution is regular there. At the crossing, Eq. (16) stops determining y_2 and instead imposes the constraint

$$y_{1,i+1} - e^p y_{1,i} - h \varphi_1(p) \bar{a}_{13} \bar{y}_3 = 0.$$

The large discrete coefficients enforce this constraint automatically as the crossing is approached. Numerically, the code protects the division against an exact zero and replaces isolated, anomalously large reconstructed face values by a local average. It still needs no turning-point detection or matching condition.

Why this matrix can count modes. The matrix is not written symmetrically, but its two couplings across each active interior face have a positive product:

$$\mathbf{T}_{i,i+1} \mathbf{T}_{i+1,i} = \frac{e^{q_{i+1}+p_f}}{(h\varphi_1 \bar{a}_{12})_f^2} > 0. \quad (19)$$

Each connected interior block can therefore be rescaled into symmetric form. A boundary condition may set an end coupling to zero; this simply ends a block and does not change the counting argument. The factorisation provides Sturm sign information (e.g. Wilkinson 1965): after correcting for the known poles at $\bar{a}_{12} = 0$, the number of negative pivots changes whenever the trial value of σ^2 crosses an eigenvalue. Section 4 explains how SISMO uses this property to locate modes.

3.3. Boundary rows

The system is closed by two rows representing the physical boundary conditions of the original oscillation problem, expressed in the same set of variables. At the stellar surface, the Lagrangian pressure perturbation vanishes, $\delta P = 0$. Using Eq. (3), this condition reduces to

$$y_1 - y_2 + y_3 = 0 \quad (x = x_s). \quad (20)$$

The surface boundary condition is combined with the half-cell form of Eq. (16) to construct the final row of the discrete system. At the centre, regularity imposes a linear relation between y_1 and y_2 . For a grid extending to $x = 0$, this relation may be written, for example, as

$$\sigma^2 y_1 = \ell S_{2,y_2}. \quad (21)$$

In practice, the condition is imposed at the first grid point satisfying $x > 10^{-8}$. This avoids evaluating the formally singular $1/x$ terms in the coefficients at radii that are numerically indistinguishable from zero, without affecting the physical central boundary condition. Both boundary rows remain confined to the tridiagonal bandwidth and preserve the positive-product property given by Eq. (19). The Sturm-counting structure of the discrete operator is therefore retained after the boundary conditions are imposed.

4. Mode location and refinement

The calculation separates two different tasks:

1. Temporarily set the gravitational source to zero and locate the modes of the mechanical, or Cowling, problem in the requested frequency range.
2. Use each Cowling mode as the starting point for an iteration that makes the displacement and its gravitational potential mutually consistent.

The first stage answers *Where are the modes?* using only the inexpensive tridiagonal matrix. The second answers *How does the gravitational source change each mode?* and is carried out only for the modes that will be retained.

4.1. Locating the Cowling modes

For a trial value of σ^2 , SISMO factorises the tridiagonal matrix $\mathbf{T}(\sigma^2)$. Section 3.2 showed that its connected blocks can be rescaled into symmetric form. Their negative pivots therefore

provide a Sturm count. This is the main advantage over a mode-by-mode search: no individual starting frequency is required.

One correction is needed. The matrix contains $1/\bar{a}_{12}$, and the position where $\bar{a}_{12} = 0$ moves through the grid as the trial frequency changes. When this position crosses a face, one matrix pivot changes sign even though no physical eigenvalue has been crossed. At the same time, the number of faces with $\bar{a}_{12} < 0$ also changes by one. Adding the two counts removes this spurious event:

$$S(\sigma^2) = \{\text{negative pivots}\} + \{\text{faces with } \bar{a}_{12} < 0\}. \quad (22)$$

A Lamb-frequency pole leaves S unchanged, whereas a genuine eigenvalue changes it.

In practice, SISMO first chooses a frequency interval broad enough to contain the requested radial orders. The asymptotic g-mode relation is used only to set these outer limits, not to guess individual mode frequencies. The interval is sampled uniformly in period because high-order g modes are approximately equally spaced in period. The sampling is chosen so that adjacent trial points normally contain at most one mode. The code brackets an interval when S changes by an odd number; in the intended one-mode-per-interval case, this change is one. Bisection then refines the enclosed eigenfrequency to a relative precision of 10^{-12} .

A few inverse-iteration steps then recover the eigenfunction associated with each frequency. The result of the scan is an ordered set of Cowling frequencies and eigenfunctions, including the g-mode sequence and any f and low-order p modes inside the interval. Each matrix factorisation is independent and costs $O(N)$ operations, so the scan and the bisections can be distributed directly across threads.

4.2. Refinement with the gravitational source

A Cowling mode is only the starting point. In the full problem, the displacement generates a potential through Poisson's equation, while that same potential acts as a source in the mechanical equation. The solution is reached when these two statements are simultaneously true. SISMO enforces this consistency by repeating the following steps for each mode.

1. *Update the potential.* Starting from the current y_1 , the code reconstructs y_2 and solves the independent Poisson problem of Sect. 2.3. To prevent the displacement and potential from overshooting one another, the new source is mixed with the previous one:

$$y_3 \leftarrow \alpha y_3^{\text{new}} + (1 - \alpha) y_3^{\text{old}}. \quad (23)$$

The mixing factor α is configurable; the benchmark calculations use $\alpha = 0.4$.

2. *Update the mode.* At the current frequency, one factorisation of T provides two response vectors,

$$\begin{aligned} \mathbf{z} &= T^{-1} [\mathbf{r}(y_3) - T\mathbf{y}_1], \\ \mathbf{t} &= T^{-1} T' \mathbf{y}_1, \quad T' = \frac{\partial T}{\partial \sigma^2}. \end{aligned} \quad (24)$$

Here $\mathbf{z}$ corrects the mismatch in the mechanical equation if the frequency is held fixed, while $\mathbf{t}$ measures how the solution responds to a change in frequency. The derivative T' is obtained by a finite difference.

Projecting these two responses onto the current eigenfunction gives the Newton frequency correction

$$\delta\sigma^2 = \frac{\langle \mathbf{y}_1, \mathbf{z} \rangle}{\langle \mathbf{y}_1, \mathbf{t} \rangle}. \quad (25)$$

The frequency and eigenfunction are then updated together:

$$\sigma^2 \leftarrow \sigma^2 + \delta\sigma^2, \quad \mathbf{y}_1 \leftarrow \mathbf{y}_1 + \mathbf{z} - \delta\sigma^2 \mathbf{t}. \quad (26)$$

The frequency step is limited to $10^{-3}\sigma^2$ so that the iteration cannot jump to a neighbouring mode. The eigenfunction is normalised after each update. Its sign is also aligned with the preceding iterate; otherwise the arbitrary sign of an eigenfunction can make the displacement and potential alternate between two equivalent states.

3. *Test convergence.* The iteration stops when the unconstrained relative frequency correction is below $\max(10^{-7}, \epsilon_{\text{tol}})$, where ϵ_{tol} is the requested tolerance, and the potential has stabilised. If the maximum number of iterations is reached, SISMO returns the iterate with the smallest unconstrained frequency correction rather than simply returning the last one. This best-iterate rule is useful when a small residual cycle remains.

Modes are refined independently, so this stage is parallelised over modes.

4.3. The dipole closure

The Poisson solution is the sum of a particular solution and a homogeneous solution with one free amplitude A [Eq. (11)]. For $\ell \geq 2$, the surface vacuum condition determines A accurately. The dipole case, $\ell = 1$, is more sensitive: the homogeneous solution grows outwards, so a small error where the surface condition is applied can affect the potential throughout the star and noticeably shift the frequency.

For dipole modes, momentum conservation supplies a more robust condition. A uniform translation of the whole star is a zero-frequency solution, and the dipole equations therefore possess the first integral derived by Takata (2005, 2006). Defining

$$\phi = xS_2y_3, \quad \partial_x \phi = S_2y_4, \quad (27)$$

and writing $a = \xi_r$ and $\bar{p}$ for the scaled pressure perturbation, the integral is

$$J(x) = \sigma^2 x \rho a - P \bar{p} - (\sigma^2 + 2S_2)\phi + x(\sigma^2 - S_2)\partial_x \phi = 0. \quad (28)$$

For an exact dipole mode, J vanishes at every radius.

The important simplification is that J depends linearly on the free amplitude A . At grid point i , we can write

$$J_i(A) = M_i + AH_i, \quad (29)$$

where $M_i = J_i(0)$ is the value obtained from the particular solution and $H_i = J_i(1) - J_i(0)$ is the change produced by one unit of the homogeneous solution. These definitions contain the same information as the explicit expansion into particular and homogeneous terms, but make the role of A clearer.

During an intermediate iteration, the mechanical and gravitational variables are not yet fully consistent, so no single value of A generally makes every J_i exactly zero. Instead, SISMO chooses the value that minimises the residual over the whole star,

$$\sum_i w_i |M_i + AH_i|^2, \quad (30)$$

where w_i are trapezoidal integration weights. The minimum is obtained directly:

$$A = - \frac{\sum_i w_i H_i^* M_i}{\sum_i w_i |H_i|^2}. \quad (31)$$

This global closure lets all interior grid points contribute to the choice of A , instead of relying on one value at the surface. It requires only two weighted scalar products, introduces no adjustable parameter, and leaves the mechanical operator unchanged. Section 5 shows that it brings the dipole modes close to the accuracy obtained for the quadrupole modes. For rotating models, Eq. (28) must be replaced by the corresponding rotating form of the first integral.

5. Validation and benchmark

5.1. Benchmark configuration

The benchmark star is a $1.5, M_{\odot}$ main-sequence model with a convective core, evolved to an age of 638, Myr using MESA (Paxton et al. 2011). Two representations of the same equilibrium model are considered. The first is the MESA profile exported directly in OSC format, containing 1131 mesh points. This model is used as input for the GYRE reference calculation. The second is obtained from the same profile using the mesa2SISMO and intSISMO conversion sequence, which produces a remeshed model containing 24,992 points. This second model is used as the common input for both OSC and the present code. The comparison with OSC therefore isolates differences arising from the oscillation solvers, since both calculations use exactly the same equilibrium structure and numerical mesh. By contrast, the comparison with GYRE necessarily includes the effects introduced by the interpolation and remeshing of the stellar model. We estimate this contribution independently from the OSC–GYRE comparison and find a characteristic relative-frequency difference of approximately 2.2×10^{-3} . This value defines the model-dependent accuracy floor of the three-code comparison. The calculations cover harmonic degrees $\ell = 1$ and 2 and radial orders $|n| = 1, \dots, 100$. The final comparison contains 198 successfully matched modes, in addition to the fundamental and low-order pressure modes included within the frequency interval of the Sturm scan. OSC is run using its mode-by-mode grid-refinement option, with `oscQI=2`, together with its dedicated g-mode scanning procedure. GYRE version 8.1 is run in adiabatic mode using its default remeshing parameters. Modes are matched between the different codes according to nearest frequency. A matching procedure based on radial-order labels cannot be used over the complete dipole branch because the $\ell = 1$ labels reported by OSC begin to repeat above approximately $|n| \approx 53$.

5.2. Agreement with the coupled solvers

Table 1 and Figs. 1–2 summarise the accuracy of the computed frequencies. Three main results emerge from these comparisons.

Differences between the solvers follow a smooth discretisation trend. When the present code and OSC are applied to the same stellar model, the frequency differences remain small and vary smoothly with radial order. For $\ell = 2$, they increase from approximately 10^{-6} around $|n| \approx 15$ to approximately 10^{-4} at $|n| = 100$ (Fig. 1). This trend is consistent with the accumulated difference between the exponentially fitted staggered discretisation used here and the finite-difference scheme applied by OSC on mode-dependent refined grids. As expected for methods of comparable order, the discrepancy increases with the radial wavenumber of the mode. The absence of isolated outliers is also noteworthy. All modes follow the same continuous trend, indi-

cating that the Sturm scan identifies the spectrum consistently and that the subsequent fixed-point refinement converges uniformly across the selected range of radial orders.

The dipole and quadrupole branches reach comparable accuracy. With the first-integral closure applied to the Poisson problem, the $\ell = 1$ frequency differences exhibit essentially the same behaviour as those obtained for $\ell = 2$. The largest dipole discrepancy, 2.1×10^{-3} , occurs for the mode immediately adjacent to the fundamental mode. This mode also has the largest Cowling shift in the benchmark set, approximately 5.5%. For this least favourable mode, the remaining discrepancy corresponds to approximately 4% of the Cowling shift, implying that the global closure removes about 96% of the source-related error. By contrast, when the homogeneous amplitude of the potential is determined solely from the surface matching condition, the dipole errors follow an envelope approximately equal to 1.3 times the local Cowling shift. This envelope decreases roughly as $n^{-2.4}$ but reaches several percent for the lowest-order modes. The use of Takata’s first integral is therefore essential for obtaining comparable accuracy for the dipole and quadrupole branches.

The fundamental mode is located by the same counting procedure. The fundamental mode requires no dedicated search strategy. In the present formulation, it is identified as the next change in the corrected Sturm count above the g-mode sequence and is subsequently refined using the same fixed-point iteration as all other modes. Its frequency agrees with the OSC value to within 7×10^{-4} . This represents an improvement over the response-based scans used in the first implementation of the code, which were unable to identify the fundamental mode reliably because a probe constructed from a g-mode-like eigenfunction produces little resonant response at the f-mode frequency.

5.3. Anatomy of the error along the branch

The comparison against the coupled solver on the same model cleanly separates the three error sources of the method:

- *Discretisation* dominates for $|n| \gtrsim 15$: the smooth curve of Fig. 1, growing with wavenumber, identical in shape for both degrees. It is controlled by the grid (here, the native model grid) and would respond to Richardson extrapolation or higher-order fitting if smaller residuals were required.
- *Residual dipole source error* dominates the lowest dipole orders: the $\lesssim 2 \times 10^{-3}$ remnants of the $\approx 5\%$ Cowling shifts, i.e. the part of the potential not captured by the least-squares first integral with the converged (but not exact) mechanical source.
- *Fixed-point truncation* is negligible: modes converge to the raw Newton criterion of 10^{-8} , and the best-iterate rule bounds the effect of any residual limit cycle below the discretisation error.

5.4. Computational cost

The timings are summarised in Table 2. Relative to the coupled reference calculation performed on the same stellar model, the approximately order-of-magnitude reduction in wall-clock time results from three main properties of the numerical formulation.

First, each evaluation of the mechanical operator requires only the assembly and factorisation of a scalar tridiagonal

Table 1. Relative frequency differences of the present code over the full benchmark set ($\ell = 1, 2$, $|n| = 1 \dots 100$). The OSC comparison uses the identical 25 000-point model (pure solver difference); the GYRE comparison is dominated by the model re-interpolation floor, measured independently as OSC–GYRE.

	vs OSC (same model)		vs GYRE	
	$\ell = 1$	$\ell = 2$	$\ell = 1$	$\ell = 2$
median $ \delta_{\text{rel}} $	1.9×10^{-5}	1.8×10^{-5}	2.16×10^{-3}	2.17×10^{-3}
fraction $< 10^{-3}$	99%	100%	–	–
fraction $< 10^{-2}$	100%	100%	100%	100%
maximum	2.1×10^{-3}	1.0×10^{-3}	–	–
(OSC vs GYRE)			2.18×10^{-3}	2.16×10^{-3}

1M5 g-modes: SISMO vs OSC (same intrnad model -- pure solver difference)

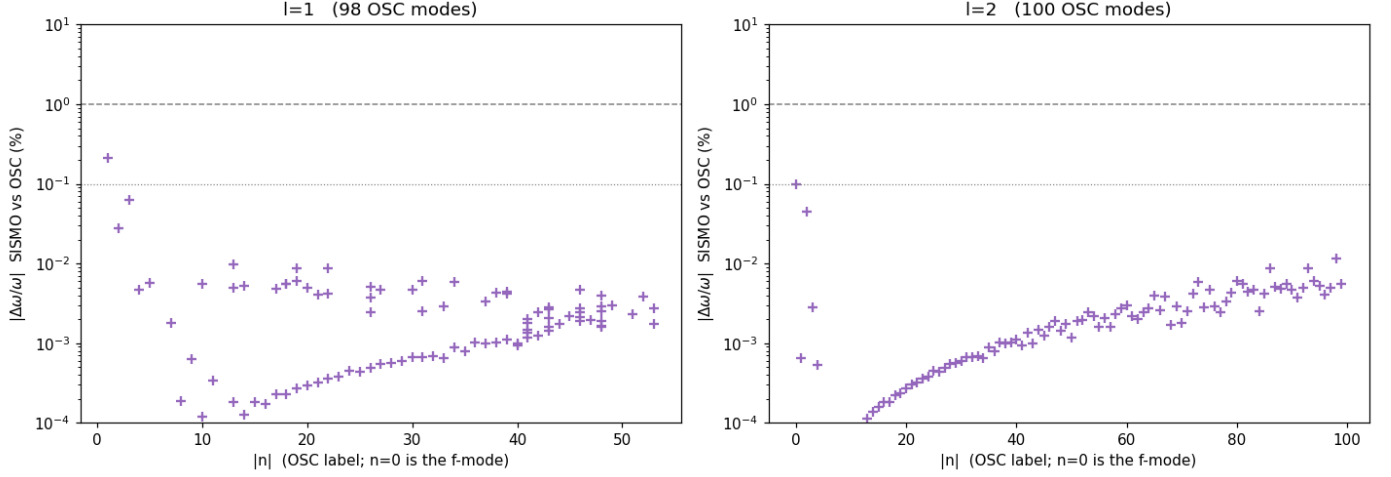


Fig. 1. Relative frequency differences between the present split-Poisson code and OSC, computed using the same 24,992-point stellar model, as a function of the radial-order label reported by OSC. The fundamental mode is labelled $n = 0$. The left and right panels show the results for $\ell = 1$ and $\ell = 2$, respectively. The smooth increase towards high radial order reflects the difference between the two discretisation schemes. All modes, including the fundamental mode, agree to better than 2.1×10^{-3} .

Table 2. Wall-clock and CPU times for the complete benchmark ($\ell = 1, 2$, $|n| = 1 \dots 100$) on one 18-core workstation. GYRE runs on its own 1131-point input model with internal remeshing and is listed for context; OSC, the first-generation code, and the present code all process the identical 24 992-point model.

Code	wall	CPU	modes
GYRE 8.1 (own grid)	9.4 s	71 s	200
OSC (scang, oscQI=2)	72.5 s	1095 s	202
First-generation split code	72.7 s	905 s	204
Present code	6.6 s	106 s	216

matrix. Both operations scale as $O(N)$ and have a small numerical prefactor, in contrast to the factorisation of the 4×4 block-banded system used in the fully coupled formulation. This low cost makes the direct Sturm scan practical: approximately 10^4 factorisations of the complete 24,992-point matrix are performed in about two seconds.

Second, the Sturm scan eliminates the need for mode-dependent initial frequency estimates and associated seeding procedures. The matrix factorisations at the trial frequencies, the bisections of the identified eigenvalue intervals, and the subsequent mode refinements are mutually independent. These operations can therefore be distributed directly across threads with little communication overhead. For the complete benchmark, the

wall-clock time decreases from approximately 104,s in serial execution to 6.6,s when using 18 threads. Third, the calculation is performed on the original model grid throughout. No multigrid procedure, mode-dependent mesh refinement, or intermediate remeshing is required. In addition to reducing the computational overhead, this avoids introducing interpolation errors associated with transferring the equilibrium structure or the eigenfunctions between different grids. The memory requirement consists of a small number of $O(N)$ work arrays for each thread. The complete implementation contains approximately 2500 lines of source code.

6. Summary and outlook

We have shown that a fully inhomogeneous (split) treatment of the gravitational potential reproduces coupled adiabatic oscillation calculations at the level of the discretisation error – median 2×10^{-5} , maximum 2×10^{-3} relative against OSC on the identical model over the complete $\ell = 1, 2$, $|n| \leq 100$ spectrum including the fundamental – at roughly one tenth of the computational cost. The three ingredients that make this possible are:

1. an exact reduction of the mechanical pair to one scalar second-order equation, discretised by a cell-wise exponentially fitted box scheme whose tridiagonal matrix has positive off-diagonal products;
2. mode location by a pole-corrected Sturm parity scan of that matrix, which locates gravity, fundamental, and pressure

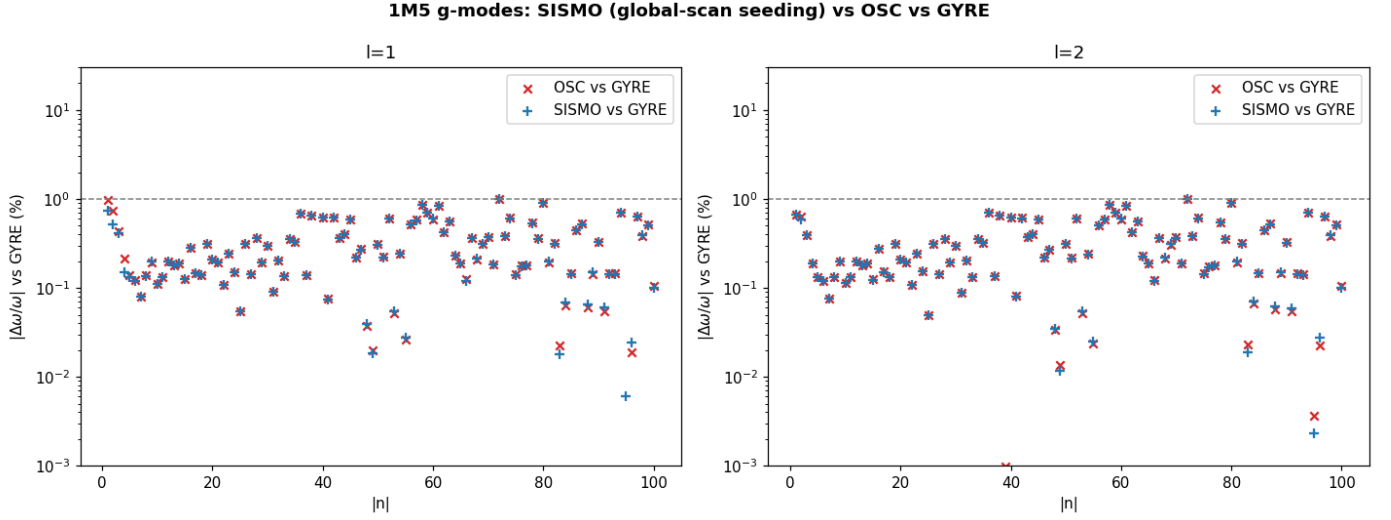


Fig. 2. Relative frequency differences with respect to GYRE for $\ell = 1$ (left) and $\ell = 2$ (right). Crosses show the OSC–GYRE comparison, while plus signs show the differences between the present code and GYRE. The two sets of differences follow the same characteristic level of approximately 2×10^{-3} , which is attributed primarily to the interpolation and remeshing of the equilibrium model.

modes without mode-by-mode asymptotic seeding, provided that the period grid resolves adjacent modes;

3. for the dipole, a parameter-free closure of the independent Poisson problem by least-squares enforcement of Takata’s first integral, replacing the surface matching condition whose globally propagated amplitude error was identified as the origin of the classical split-scheme dipole bias.

The eigenvalue operator never contains the potential; the architecture is therefore ready for the programme that motivated it: the perturbative and non-perturbative introduction of rotational (Coriolis) source terms computed from the current eigenfunction, using the same frozen-source iteration, the same Sturm location on the non-rotating operator, and – for the dipole – the rotating generalisation of the first integral.

Acknowledgements. ...

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